

A Proof of Conjecture 00000000049

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Claude (Opus 5)[†]

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Abstract

Conjecture 00000000049 asserts, of point sets in the plane all of whose pairwise distances are prime, that the maximal size is 4 and that a set of size 4 exists. Both hold. A set of size 4 is $\{0, 2, 5, 7\}$ on a line. For the bound we show that among the six distances of any such quadruple *exactly two equal 2*: the six squared distances satisfy the Gram relation of four coplanar points, each is 4 or is 1 modulo 8, and a check of the 64 resulting patterns leaves only a perfect matching, a triangle, or all six pairs — the last two being excluded by the exact value 128 and by a computation modulo 32. Five points would then give $3e = 5 \cdot 2$ for the number e of pairs at distance 2, which is impossible. The whole conjecture, both parts, is formalized in Lean 4 against Mathlib.

1 Statement and conventions

We identify the Euclidean plane with $\mathbb{C}$. Call a finite set $S \subset \mathbb{C}$ a *prime-distance set* if $|P - Q|$ is a prime integer for all distinct $P, Q \in S$. Conjecture 00000000049 asserts

- (a) every prime-distance set has at most 4 elements;
- (b) some prime-distance set has 4 elements.

Throughout, “prime” means a rational prime $2, 3, 5, 7, \dots$; note that 2 is allowed, and it is the only even one. That is what makes the problem differ from the classical question about odd integral distances, and, as it turns out, the distance 2 is what the whole proof is about.

2 Part (b)

Proposition 1. $S = \{0, 2, 5, 7\} \subset \mathbb{R} \subset \mathbb{C}$ is a prime-distance set of size 4.

Proof. The six distances are

$$|0 - 2| = 2, \quad |0 - 5| = 5, \quad |0 - 7| = 7, \quad |2 - 5| = 3, \quad |2 - 7| = 5, \quad |5 - 7| = 2,$$

all prime. □

3 Part (a)

3.1 The Gram relation

Let P_0, P_1, P_2, P_3 be four points of the plane, write $v_i = P_i - P_0$, and put

$$s_i = |P_0 P_i|^2, \quad t_{ij} = |P_i P_j|^2.$$

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The Gram matrix $G = (v_i \cdot v_j)_{1 \leq i, j \leq 3}$ is singular, because v_1, v_2, v_3 lie in a 2-dimensional space. Since $2 v_i \cdot v_j = s_i + s_j - t_{ij}$ for $i \neq j$ and $2 v_i \cdot v_i = 2s_i$, setting

$$x_{ij} = s_i + s_j - t_{ij}$$

and expanding $\det(2G) = 0$ gives, after dividing by 2,

$$E := 4s_1s_2s_3 + x_{12}x_{13}x_{23} - s_1x_{23}^2 - s_2x_{13}^2 - s_3x_{12}^2 = 0. \quad (1)$$

As a polynomial identity in the coordinates of the four points this is the statement that the Gram determinant of three coplanar vectors vanishes; it is $\pm \frac{1}{2}$ times the Cayley–Menger determinant of the quadruple [2], and in particular it is symmetric under relabelling the points.

Everything below is an arithmetic consequence of (1). Note that the six squared distances of a prime-distance quadruple are integers, and

$$p^2 \equiv \begin{cases} 4 & (\text{mod } 8), \quad p = 2, \\ 1 & (\text{mod } 8), \quad p \text{ an odd prime,} \end{cases} \quad (2)$$

since an odd square is $\equiv 1 \pmod{8}$.

3.2 Which pairs can be at distance two

Let S be a prime-distance set of size 4 and let H be the graph on S whose edges are the pairs at distance exactly 2. By (2) the residue of E modulo 8 depends only on H , so it can be computed once for each of the eleven graphs on four vertices.

Lemma 2. *H is a perfect matching (two disjoint edges), a triangle together with an isolated vertex, or the complete graph.*

Proof. Substituting 4 for the squared distances of the edges of H and 1 for the others in (1) and reducing modulo 8 gives, for the eleven isomorphism types on four vertices:

H	edges	$E \pmod{8}$
empty	0	2
one edge	1	4
two adjacent edges	2	7
perfect matching	2	0
path P_4	3	7
star $K_{1,3}$	3	3
triangle + isolated vertex	3	0
4-cycle	4	6
triangle + pendant edge	4	4
K_4 minus an edge	5	4
K_4	6	0

All but the three boldface types have $E \not\equiv 0 \pmod{8}$, contradicting (1). (The computation is a finite check over the $2^6 = 64$ labelled patterns; it is the Lean lemma `pattern_mod8`.) $\square$

Lemma 3. *H is not the complete graph.*

Proof. All six squared distances would equal 4, and then $E = 4 \cdot 64 + 64 - 3 \cdot 4 \cdot 16 = 128 \neq 0$. $\square$

Lemma 4. *H is not a triangle together with an isolated vertex.*

Proof. Take the isolated vertex as the base point P_0 , so that $t_{12} = t_{13} = t_{23} = 4$ while $s_1 = p^2$, $s_2 = q^2$, $s_3 = r^2$ for odd primes p, q, r . The square of an odd number is 1, 9, 17 or 25 modulo 32, and for each of the $4^3 = 64$ choices (1) gives

$$E \equiv 16 \pmod{32},$$

never 0. (Again a finite check: the Lean lemma `tri_mod32`, which covers the four positions of the isolated vertex.) $\square$

Theorem 5. *In a prime-distance set of size 4, exactly two of the six distances equal 2, and those two pairs are disjoint.*

Proof. Lemmas 2, 3 and 4. $\square$

3.3 No five points

Theorem 6. *Every prime-distance set has at most 4 elements.*

Proof. Suppose S is a prime-distance set with $|S| \geq 5$ and pick five points $Q_1, \dots, Q_5$ of it. Let e be the number of pairs among them at distance 2. Each of the five 4-element subsets is a prime-distance set, so by Theorem 5 it contains exactly two such pairs; and each pair lies in exactly $\binom{3}{2} = 3$ of those subsets. Counting the incidences between pairs at distance 2 and 4-element subsets in two ways,

$$3e = 5 \cdot 2 = 10,$$

which has no solution in integers. $\square$

Corollary 7. *Conjecture 00000000049 holds.*

Proof. Theorem 6 and Proposition 1. $\square$

Remark 8. The bound needs no geometry beyond (1): no case analysis on configurations, no coordinates, and — in contrast with the usual route through the Erdős–Anning theorem [1] — no collinearity. Collinearity is nevertheless true here, and Appendix A proves it, together with the fact that $\{0, 2, 5, 7\}$ is the *only* quadruple up to isometry.

4 Formal verification

The accompanying Lean 4 project proves the conjecture in full: the final theorem is

`conjecture_holds : ConjectureHolds,`

where `ConjectureHolds` is the conjecture itself, stated in the formalization as

$(\forall n (P : \text{Fin } n \rightarrow \mathbb{C}), \text{IsPrimeDistanceSet } P \rightarrow n \leq 4) \wedge (\exists P : \text{Fin } 4 \rightarrow \mathbb{C}, \text{IsPrimeDistanceSet } P).$

- `IsPrimeDistanceSet`, `ConjectureHolds` — the notion and the conjecture, with the plane modeled as $\mathbb{C}$ and primality Mathlib’s `Nat.Prime`;
- `witness`, `dTable_prime`, `conjecture_part_b` — Proposition 1;
- `gram`, `gram_eq_zero`, `gram_dist`, `gram_int` — the relation (1). The identity itself is proved by `ring` on the coordinates of the three difference vectors;
- `pattern_mod8` — Lemma 2, as a `decide` over the 64 labelled patterns;
- `tri_mod32`, `odd_sq_mod32` — Lemma 4, likewise by `decide`;

- `two_of_six`, `two_of_four` — Theorem 5, in the counting form used next: the six indicators of “distance = 2” sum to 2;
- `no_five`, `card_le_four` — Theorem 6; the double count is discharged by `omega` from the five equations;
- `conjecture_holds` — the conjecture;
- `gaps_of_four_collinear` — the arithmetic half of Appendix A (Lemma 10).

Scope. Parts (a) and (b) are both formalized, with no gap: the Lean development contains no `sorry`, no `admit` and no `native_decide`, and `#print axioms conjecture_holds` reports exactly the three standard axioms `propext`, `Classical.choice` and `Quot.sound`. What is *not* formalized is the supplementary Appendix A: its geometric step (Lemma 9) is a coordinate computation that would need a rigid-motion normalization in Lean, and nothing in the conjecture depends on it. The project builds with `lake build on leanprover/lean4:v4.33.1` with `Mathlib v4.33.1`.

5 Computational corroboration

Four points are planar exactly when their 5×5 Cayley–Menger determinant vanishes; with prime squared distances this is an exact integer test. Over all 6-tuples of primes below 120 satisfying the triangle inequalities — about $4.4 \cdot 10^6$ configurations — the only solutions are degenerate, and every one of them is $\{0, 2, 5, 7\}$ up to relabelling and reflection. This agrees with Theorem 5 and with Appendix A. The search is corroboration, not part of the proof.

A The quadruples are collinear, and there is only one

Theorem 5 says that a prime-distance quadruple has exactly two distances equal to 2, on disjoint pairs. That is already enough for the conjecture. With one coordinate computation more, the quadruples can be determined completely. Nothing in this appendix is used above, and it is not formalized apart from Lemma 10.

Lemma 9. *Let $\{A, B, C, D\}$ be a prime-distance set of size 4 with $|AB| = |CD| = 2$ and the other four distances odd. Then the four points are collinear.*

Proof. Place $A = (-1, 0)$, $B = (1, 0)$. For any point $P = (x, y)$ with $|AP|, |BP|$ odd,

$$x = \frac{|AP|^2 - |BP|^2}{4}$$

is an even integer, because $|AP|^2 - |BP|^2 \equiv 0 \pmod{8}$; and $y^2 = |AP|^2 - (x+1)^2 \equiv 1 - 1 = 0 \pmod{8}$. Apply this to $C = (x_C, y_C)$ and $D = (x_D, y_D)$.

From $|CD|^2 = 4$ and $x_C - x_D$ even we get $(x_C - x_D)^2 \in \{0, 4\}$.

If $x_C = x_D$ then $(y_C - y_D)^2 = 4$, so $2y_C y_D = y_C^2 + y_D^2 - 4 \equiv 4 \pmod{8}$; thus $y_C y_D$ is an integer $\equiv 2 \pmod{4}$ and $(y_C y_D)^2 \equiv 4 \pmod{32}$. But $(y_C y_D)^2 = y_C^2 y_D^2 \equiv 0 \pmod{64}$. Contradiction.

So $|x_C - x_D| = 2$ and $y_C = y_D =: y$. Relabelling, write $C = (x, y)$, $D = (x+2, y)$ and put $p = |AC|$, $q = |AD|$, $r = |BC|$; then $|BD| = |AC| = p$ and

$$q^2 + r^2 = 2p^2 + 8, \quad x = \frac{p^2 - r^2}{4}, \quad y^2 = p^2 - (x+1)^2.$$

Now $y^2 \geq 0$ forces $(p-2)^2 \leq r^2 \leq (p+2)^2$, i.e. $p-2 \leq r \leq p+2$; as p, r are odd, $r \in \{p-2, p, p+2\}$. If $r = p$ then $q^2 = p^2 + 8$, so $(q-p)(q+p) = 8$ with both factors even,

forcing $q - p = 2$, $q + p = 4$ and $p = 1$, not prime. Hence $r = p \pm 2$, which gives $x + 1 = \pm p$ and $y^2 = 0$: the four points lie on the x -axis. $\square$

Lemma 10. *Let $x_0 < x_1 < x_2 < x_3$ be four points on a line with all six pairwise distances prime, and let $g_i = x_i - x_{i-1}$. Then $(g_1, g_2, g_3) = (2, 3, 2)$.*

Proof. Each g_i is prime, and so is each sum of consecutive ones. Two adjacent gaps cannot both be odd, since their sum would be even and at least 6; and they cannot both be 2, since 4 is not prime. So of each adjacent pair exactly one equals 2, which leaves the patterns (2, odd, 2) and (odd, 2, odd). In the second the total span $g_1 + g_2 + g_3$ is even and at least 8, hence not prime; so $g_1 = g_3 = 2$ and $q := g_2$ is odd.

Now q , $q + 2$ and $q + 4$ are all prime — they are $|x_1x_2|$, $|x_1x_3|$ and $|x_0x_3|$ — and one of any three integers in arithmetic progression of difference 2 is divisible by 3. A prime divisible by 3 equals 3; since $q \geq 3$ the values $q + 2$ and $q + 4$ exceed 3, so the one divisible by 3 is q itself, and $q = 3$. $\square$

Theorem 11. *Every prime-distance set of size 4 is collinear, and is $\{0, 2, 5, 7\}$ up to an isometry of the plane.*

Proof. By Theorem 5 the two distances equal to 2 lie on disjoint pairs, so Lemma 9 applies and the set is collinear; by Lemma 10 its consecutive gaps are 2, 3, 2, so after an isometry of the line carrying it into $\mathbb{R}$ it is $\{0, 2, 5, 7\}$. $\square$

Remark 12. The same gap analysis reproves, in one line, that no five points on a line have prime pairwise distances: the four gaps would have to alternate between 2 and an odd prime, making the span 4 plus two odd primes — even and at least 10.

References

- [1] N. H. Anning, P. Erdős, *Integral distances*, Bull. Amer. Math. Soc. **51** (1945), 598–600.
- [2] L. M. Blumenthal, *Theory and Applications of Distance Geometry*, Clarendon Press, Oxford, 1953.